	7.4. MATERIAL HANDLING AND STORAGE	
AzurESky	Document no: ASMF 7.4 (also covers 7.14)	
Compiled by:	Sifiso Nyambi - Laboratory Manager	
Approved by:	Nonhlanhla Msibi - Operations Manager	

Material Handling and Storage

Objective

To maintain the **identity, quality, stability, and traceability** of all materials (raw, in-process, and final) through appropriate handling and storage procedures within the laboratory.

1. Material Identification and Traceability

- Every **material unit** (raw, processed, final RM) must have a **unique batch ID**.
- Use clearly labelled containers with:
- Material name
- Batch/lot number
- Status (e.g. Quarantine, Approved, Rejected)
- Storage conditions
- Expiry/retest date

2. Handling Procedures

- Materials must be handled in a way that prevents contamination, mix-ups, or degradation.
- SOPs must exist for:
- Receiving and inspecting raw materials
- Handling in-process materials (e.g., sub-samples, blended batches)
- Transferring materials between areas or containers
- Using dedicated, clean containers and tools during handling
- Ensuring segregation of incompatible or unstable materials (if applicable)

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07/2025	00	07/2028

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- Since coal is a relatively stable and low-volatility material, the effects of transportation on its chemical composition are generally minimal, provided that samples are properly contained. However, to prevent moisture loss, contamination, or misidentification, it is essential that all sample bags are securely sealed and clearly labelled to preserve the integrity and traceability of the sample

3. Storage Area Requirements

- Storage conditions must match the **stability requirements** of the material:
- Temperature and humidity controlled as necessary (Temperature ≤20 °C, RH 25-60%)
- Secure from unauthorized access
- Clean, dry, and organized by batch or same category
- Implement:
- **Environmental monitoring logs** (continuous monitoring, Mon & Fri.) See document **AS018TF**

4. Segregation and quarantine control

- Separate zones or shelves for:

- Raw material – Preparation Area
- Approved material – Ready for use
- In-process material – Awaiting testing or validation (Homogeneity drum)
- Final PT/RM – Ready for distribution

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5. Inventory Control System

- Maintain a live **Material Inventory Register** including:
- Batch number
- Quantity in stock
- Date received
- Supplier/source
- Assigned expiration or review date
- Apply **FIFO (First in First Out)** or **FEFO (First Expired First Out)** logic to material use

6. Documentation Requirements

The following records must be kept and regularly reviewed:

Record	Purpose
Material Receipt	Logs incoming material, supplier, inspection result, and quarantine status
Storage Condition Monitoring Log	Record temperature/humidity and any excursions
Inventory Register	Track samples by batch, quantity, and status
Non-conformance and deviation reports	Document handling or storage errors and corrective actions
Cleaning & Maintenance Records	Logs for cleaning the storage area to prevent contamination

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7. Staff Training

All personnel handling material must be:

- Trained in SOPs for material handling and storage

- Aware of contamination risks and segregation practices
- Authorised (access-controlled areas should be restricted to trained staff)

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